2.3.2.2 n8n Marketing Boost Mechanism

n8n is an open-source workflow automation platform that connects applications, databases and AI services through a series of connected steps called nodes [1]. Because the steps are arranged visually, repetitive integration tasks need little custom code [1]. In a small-scale business case study, an automated n8n workflow showed no errors, while the same process done manually had a 5% error rate. The authors describe this result as preliminary [2]. Social media gives farmers and small sellers an extra channel to reach buyers, but many lack a clear posting strategy [3].

In Agrivive, n8n coordinates the promotion steps for a surplus vegetable listing. It receives a job from the backend, fetches the listing's current facts, asks an AI model to draft a caption, and returns the caption to the backend. The backend then validates the post and publishes it to Facebook and Instagram. Each post gives households, carinderias, restaurants and resellers another chance to see a perishable listing before it spoils. Follow-up price drops add a second chance. Lowering the price of perishables as they age is a recognized way to reduce retail food waste [4], [5]. Increased visibility does not guarantee that buyers will view, reserve or purchase a listing.

Division of Responsibility

Component	Does	Does not do
ElysiaJS backend	Checks eligibility, calculates prices, removes links from captions, adds the product link, saves post records, posts to Facebook and Instagram, tracks the 6-hour timing	Write captions
n8n	Receives the job, fetches fresh listing data, calls the AI model, passes the caption back	Set prices or decide eligibility
openai/gpt-oss-20b (via Groq)	Drafts caption text only	Decide eligibility, prices or visibility
PostgreSQL	Stores listings, boost records and post history	n/a

Workflow

Stage 1: First post

1. **Seller triggers a Boost.** The seller taps Boost on a listing. The backend checks that the seller is not paused and has a fault rate below 15%; the listing is active, marketable, in stock, and has a photo; no first post was made for this product in the last 6 hours; and no boost already exists for this listing cycle. The backend then saves a boost record in PostgreSQL with the status *queued*.
2. **The worker sends the job to n8n.** A separate worker process runs every 60 seconds. It finds due boost records, claims each with a short lease (15 seconds) so no other worker picks it up, and sends only the boost ID and reason to the n8n webhook. The request carries a service token over HTTPS. Failed deliveries retry up to 5 times with exponential backoff, capped at 30 minutes.

3. **n8n fetches the current listing facts.** It calls a protected backend endpoint for the product name, unit, price, quantity, image URL, product link and shop name. It uses no cached copy.
4. **n8n drafts the caption.** It sends only the product name, unit, price and shop name to openai/gpt-oss-20b through Groq. The model returns JSON with a single *caption* field. The prompt tells it to use only the given facts, not to change any number, not to write a link, and not to use urgent wording. If the response is missing or invalid, a fixed default caption is used.
5. **The backend publishes the post.** It removes any link the model wrote; replaces the caption with a template if the current price is missing; adds the product link with tracking parameters; saves a pending post record before anything is sent, so a repeated request cannot post twice; posts the photo and caption to Facebook, then Instagram, saving a separate result for each; sets the follow-up due time to 6 hours after the first post; and notifies the seller that the boost has started.

Stage 2: Follow-up price drops

n8n does not keep a workflow waiting for 6 hours. The timer lives in PostgreSQL.

1. **The backend schedules the next round.** When the follow-up is due, the worker checks whether a drop is still possible. The listing must be active, marketable and in stock. The seller must have turned on auto-post and price drops, and the price must be above the seller's final price. If not, the rule ends and n8n is not called. If a drop is possible, the worker sends a follow-up job to n8n, retrying every 15 minutes, up to 5 attempts.
2. **n8n asks the backend to cut the price.** The backend locks the product record, finds the price in the most recent successful post, and skips the cut if the price is already below that post, so retries are safe. Otherwise it applies the cut. If no cut is made, the rule ends and n8n posts nothing.
3. **Second post.** If the price was cut, n8n writes a flash-sale caption with the discount percentage, old price and new price. The backend posts it to Facebook and Instagram, notifies the seller that the price was lowered, and sets the next check 6 hours ahead.
4. **Stop conditions.** The rule ends, and n8n is not called, when the listing is sold out, archived or paused. It also ends when the price is at the final price, when the seller turns price drops off, or when the listing starts a new cycle.

The AI model drafts text only. It does not decide eligibility, calculate prices, or assign visibility tiers.

Formula

The seller sets a fixed peso drop per round, D, and a final price, P_{min}. ElysiaJS performs the calculation and PostgreSQL stores the updated price. All values are computed in centavos, so no rounding error occurs.

$$P_{new} = max(P_{current} - D , P_{min})$$

Where P_{new} is the updated price, $P_{current}$ is the current price, D is the peso drop per round, and P_{min} is the seller's final price. The cut applies only when the listing is eligible, $D > 0$, a final price is set, and $P_{current} > P_{min}$.

The discount percentage shown in the caption is calculated from the actual cut:

$$Discount\ (\%) = round(\ (\ P_{before} - P_{after}\ )\ /\ P_{before} \times 100\ )$$

The fixed peso drop and the 6-hour interval are Agrivive prototype settings, not figures from the literature. Markdown pricing of perishables is supported in [4], [5].

Example Scenario

A seller lists tomatoes at ₱80/kg. They set D = ₱10 and P_{min} = ₱60, with auto-post and price drops on.

Time	Event	Price	Result
8:00	Seller taps Boost. The worker sends the job, and n8n drafts the caption from the listing facts.	₱80	First post on Facebook and Instagram. Follow-up due at 14:00.
14:00	Follow-up due. The backend cuts the price: max(80 − 10, 60) = 70. Discount = round(10/80 × 100) = 13%.	₱70	Flash-sale post ("13% off, was ₱80, now ₱70"). Next check at 20:00.
20:00	Cut: max(70 − 10, 60) = 60. Discount = round(10/70 × 100) = 14%.	₱60	Flash-sale post. The price is now at P_{min}.
02:00	Price equals P_{min}.	₱60	The rule ends and n8n is not called.

If the listing sells out at any point, the rule ends at the next check and nothing more is posted. If a repeated request arrives after a successful cut, the backend sees the price is already below the last post and does not cut or post again.

Records and Failures

The backend records each promotion step in PostgreSQL: the boost identifier, listing cycle, stage (first or second post), caption, platform results and timestamps. It also stores the Facebook post identifier and the Instagram post link. If a Facebook post fails, its record is marked failed and the seller is notified. Facebook and Instagram are the external channels that show the posts and direct buyers to the Agrivive product page.

References

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